

AIP-Speech: Auditing Speech LLM Robustness to Background Interference

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Abstract

Speech Large Language Models (Speech LLMs) are increasingly deployed in real-world acoustic environments, yet it remains unclear which measurable properties of background sounds best explain performance degradation. We present AIP-Speech, an inference-only benchmark that evaluates Speech LLM robustness under controlled mixtures of real-world background recordings. Rather than comparing robustness across background categories alone, we characterize each recording using interpretable acoustic descriptors and examine their associations with degradation. Across five open Speech LLMs, we find that simple spectral descriptors—including Zero Crossing Rate, Spectral Centroid, and Spectral Rolloff—consistently exhibit strong associations with degradation. We further quantify robustness disparities across accents and evaluate whether inference-time prompting improves resilience under adverse acoustic conditions. AIP-Speech provides a reproducible benchmark and descriptor-annotated background corpus for analyzing Speech LLM robustness.

1 Introduction

Speech large language models integrate speech encoders with pretrained language models to perform tasks such as transcription, keyword spotting, and spoken question answering directly from raw audio (Chu et al., 2024; Xu et al., 2025; Abouelenin et al., 2025). Modern Speech LLMs adopt an end-to-end architecture that unifies speech perception and language reasoning within a single model (Peng et al., 2025). Speech, however, is rarely encountered in isolation.

How Speech LLMs behave under background interference remains poorly understood. Existing benchmarks evaluate Speech LLMs under environmental variation (Chen et al., 2026), speaker-side phenomena such as speech disfluency (Liu et al.,

2025), natural in-the-wild recordings (Zhang et al., 2025), or systematic perturbations (Shah et al., 2025). Broader benchmarks additionally assess trustworthiness across dimensions such as robustness, safety, fairness, and hallucination (Li et al., 2025). However, analyses of background interference typically report aggregate performance under a limited set of acoustic conditions, providing little insight into which measurable acoustic properties are associated with degradation. Moreover, it remains unclear whether background interference affects demographic groups uniformly or amplifies existing disparities across attributes such as accent and gender.

To address these gaps, we introduce **Acoustic Interference Profiling (AIP)-Speech**, an inference-only benchmark for systematically auditing Speech LLM robustness under controlled background interference¹. By holding the spoken content fixed while systematically varying the background audio, we isolate the effect of different classes of acoustic interference on Speech LLM behavior. Specifically, our work makes three contributions: (i) we introduce a reproducible benchmark and evaluation framework for auditing Speech LLMs under controlled real-world background interference;

(ii) we present descriptor-level analysis of Speech LLM robustness, relating interpretable acoustic descriptors to degradation across automatic speech recognition and keyword spotting;

(iii) we evaluate robustness disparities across demographic groups and investigate inference-time prompting under controlled acoustic interference.

2 AIP-Speech Benchmark

Figure 1 summarizes the AIP-Speech evaluation pipeline. A clean speech utterance is mixed with a controlled background recording at multiple signal-to-noise ratios (SNRs) to produce paired noisy ex-

¹<https://github.com/IvLabs/AIP-Speech>

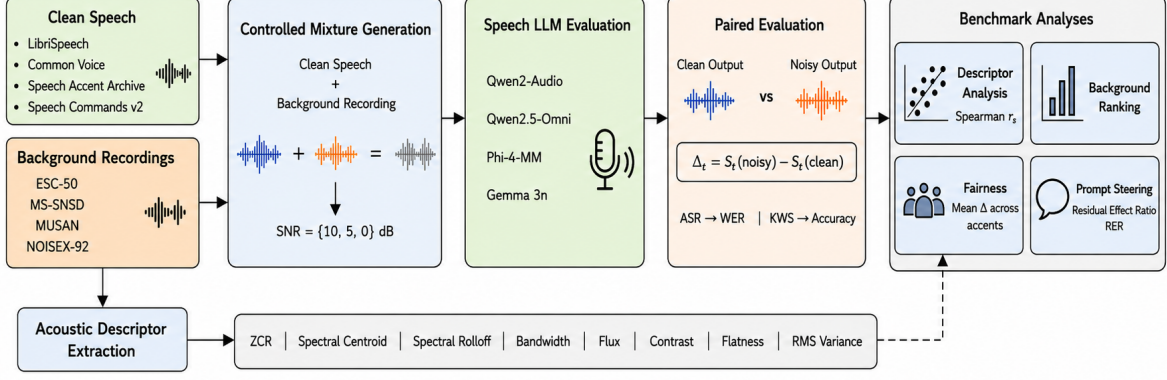


Figure 1: Overview of the AIP-Speech benchmark. Clean speech is combined with background recordings to evaluate Speech LLM robustness under controlled acoustic interference. Acoustic descriptors are extracted independently from background recordings and incorporated only during the analysis stage to explain robustness patterns.

amples. Speech LLMs are evaluated under identical inference settings, and paired clean–noisy predictions are used to compute robustness, fairness, prompting, and descriptor-based analyses. Pairing every noisy example with its clean counterpart isolates the effect of background interference from intrinsic model accuracy.

2.1 Background Audio and Descriptor Extraction

AIP-Speech uses a fixed set of 20 publicly available real-world background recordings, ensuring reproducibility while capturing diverse acoustic conditions. Recordings are drawn from ESC-50 (Piczak, 2015), MS-SNSD (Reddy et al., 2019), NOISEX-92 (Varga and Steeneken, 1993), and MUSAN (Snyder et al., 2015). Each clean speech utterance is combined with a background recording at multiple target signal-to-noise ratios (SNRs) to produce deterministic noisy counterparts. All recordings are resampled and loudness-normalized prior to mixture generation. The complete list of background recording classes is provided in Appendix A.

Each background recording is further characterized using a set of standard acoustic descriptors computed directly from the waveform. Specifically, we compute spectral centroid, spectral rolloff, spectral bandwidth, spectral flatness, zero crossing rate (ZCR), spectral flux, spectral contrast, and RMS energy variance (McFee et al., 2015). These descriptors capture complementary spectral and temporal properties of the background signal and are computed once for each recording prior to evalua-

tion. Descriptor values are subsequently related to task degradation using Spearman rank correlation.

2.2 Robustness Measures

For each evaluation task t , let $S_t(\mathbf{x})$ denote the task-specific scoring function evaluated on speech input $\mathbf{x}$. Robustness is quantified as the performance degradation Δ_t between paired clean and noisy examples $(\mathbf{x}_{\text{clean}}, \mathbf{x}_{\text{noisy}})$, formulated such that positive values consistently represent a performance drop relative to the clean baseline:

$$\Delta_t = \begin{cases} S_{\text{ASR}}(\mathbf{x}_{\text{noisy}}) - S_{\text{ASR}}(\mathbf{x}_{\text{clean}}), & \text{for } t = \text{ASR} \\ S_{\text{KWS}}(\mathbf{x}_{\text{clean}}) - S_{\text{KWS}}(\mathbf{x}_{\text{noisy}}), & \text{for } t = \text{KWS} \end{cases} \quad (1)$$

where $S_{\text{ASR}}(\cdot)$ denotes the Word Error Rate (WER) and $S_{\text{KWS}}(\cdot)$ denotes classification accuracy. Consequently, Δ_{ASR} captures the absolute increase in WER, whereas Δ_{KWS} measures the drop in accuracy. KWS uses a closed-set protocol over the ten Google Speech Commands v2 keywords, where target utterances require an exact keyword match and non-target utterances are scored correct only if no target keyword is predicted.

To quantify demographic robustness, we compute the mean performance degradation for each demographic group g ,

$$\bar{\Delta}_{t,g} = \frac{1}{N_g} \sum_{i \in g} \Delta_t^{(i)}, \quad (2)$$

where N_g denotes the number of evaluation examples belonging to group g . Larger values indicate that the corresponding demographic experi-

ences greater performance degradation under background interference.

Prompt effectiveness is summarized using the Residual Effect Ratio,

$$RER = \frac{\overline{\Delta_t^{prompt}}}{\overline{\Delta_t^{baseline}}}, \quad (3)$$

where values below one indicate reduced degradation under prompting.

3 Experimental Setup

We evaluate AIP-Speech on two speech processing tasks: automatic speech recognition (ASR) and keyword spotting (KWS). The ASR evaluation comprises 180 speech utterances, including 100 from LibriSpeech (Panayotov et al., 2015) and 80 accent-labelled utterances from Mozilla Common Voice 17 (Ardila et al., 2020). Robustness across demographic groups is evaluated using these accent-labelled utterances. The KWS evaluation comprises 113 audio clips from Google Speech Commands v2 (Warden, 2018).

We evaluate five publicly available Speech LLMs: Qwen2.5-Omni-3B, Qwen2.5-Omni-7B (Xu et al., 2025), Qwen2-Audio-Instruct (Chu et al., 2024), Phi-4-Multimodal (Abouelenin et al., 2025), and Gemma 3n (Gemma Team, 2025). All models are evaluated using deterministic greedy decoding. All models are evaluated using the same background recordings at signal-to-noise ratios (SNRs) of 10, 5, and 0 dB. Complete model-wise performance across all evaluated SNR conditions is provided in Appendix G. Robustness metrics are computed from paired clean and noisy predictions. For descriptor-level analysis, degradation is averaged across the evaluated SNR levels before computing Spearman rank correlations with the corresponding acoustic descriptors. The steering prompt is applied consistently across all evaluated models, and its complete text is provided in Appendix F.

4 Results

We evaluate Speech LLM robustness under controlled acoustic interference using AIP-Speech through descriptor analysis, background ranking, fairness evaluation, and inference-time prompt steering.

4.1 Descriptor-Level Analysis

Figure 2 demonstrates that robustness degradation is not randomly distributed across background

recordings. Instead, recordings with larger descriptor values consistently tend to produce greater degradation for both ASR and KWS, despite the inherent variability expected from real-world acoustic environments. The similar trends observed across two fundamentally different speech tasks suggest that the relationship reflects a property of the background signal rather than task-specific behavior.

Figure 3 shows that spectral descriptors are strongly associated with degradation. Zero Crossing Rate (ZCR) exhibits the strongest marginal association with degradation ($r_s = 0.645$ for ASR and 0.723 for KWS), closely followed by Spectral Centroid and Spectral Rolloff, while descriptors such as Spectral Flatness, Spectral Contrast, and RMS Energy Variance show weaker associations. Moreover, the ordering of descriptor associations is consistent across ASR and KWS. To assess consistency across architectures, we compute these correlations independently for each model in Appendix C. The strongest descriptors remain positively associated with degradation across all five models, although the correlation magnitudes vary by architecture. Collectively, these results indicate that simple, low-level spectral descriptors provide informative signals for characterizing degradation.

4.2 Background Ranking

Figure 4 provides a recording-level perspective on Speech LLM robustness. The observed variability across recordings is consistent with the descriptor analysis, where Zero Crossing Rate, Spectral Centroid, and Spectral Rolloff exhibit the strongest associations with degradation across both tasks. The complete descriptor values for every background recording are provided in Appendix D.

4.3 Fairness

Figure 5 compares the robustness of Speech LLMs across accent subgroups by measuring the mean increase in Word Error Rate (ΔWER) between clean and noisy speech. Computing degradation relative to each subgroup’s clean baseline isolates the effect of background interference from differences in baseline recognition performance. The figure reveals substantial variation in robustness across both models and accent subgroups. While some models exhibit relatively uniform degradation, others show pronounced sensitivity to specific accents, indicating that background interference does not affect all

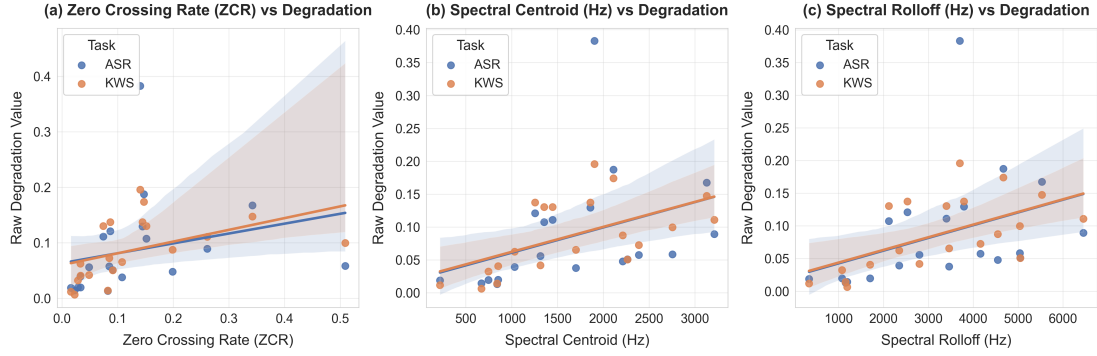


Figure 2: Relationship between three representative acoustic descriptors and task degradation. Each point corresponds to one background recording. Larger descriptor values generally correspond to greater degradation for both ASR and KWS.

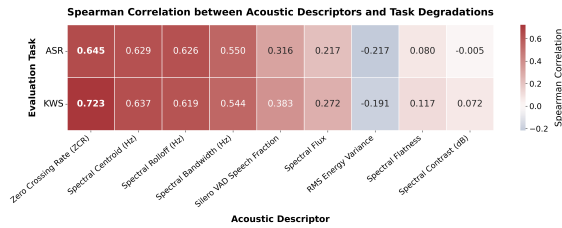


Figure 3: Spearman rank correlation between acoustic descriptors and task degradation. Spectral descriptors exhibit the strongest associations across both evaluation tasks.

demographic groups equally. Complete subgroup-wise statistics are reported in Appendix E.

4.4 Prompt-based Steering

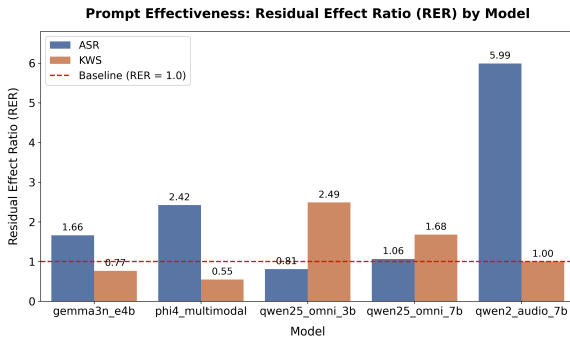


Figure 6: Residual Effect Ratio (RER) after applying inference-time prompt steering. Values below one indicate reduced degradation relative to the baseline instruction.

We evaluate whether inference-time prompt steering improves robustness without modifying model parameters. Figure 6 reports the Residual Effect Ratio (RER), which measures the degradation remaining after steering relative to the baseline in-

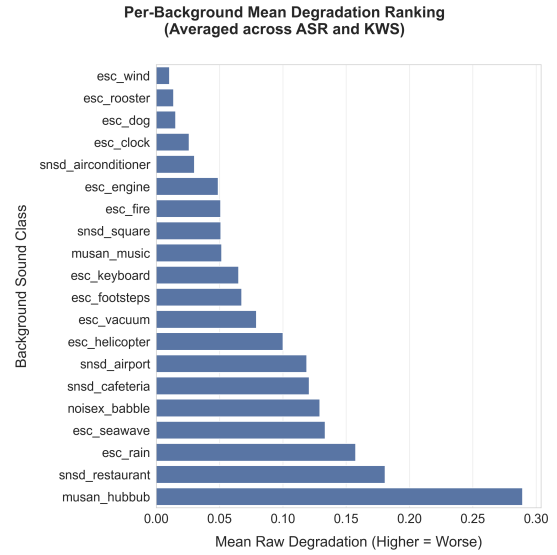


Figure 4: Mean degradation ranking of individual background recordings averaged across ASR and KWS.

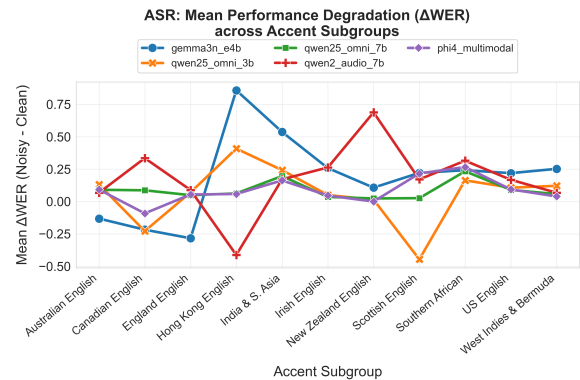


Figure 5: Mean ASR performance degradation (Δ WER) across accent subgroups under background interference. Larger values indicate greater degradation relative to the corresponding clean condition.

struction. Lower RER values indicate greater robustness gains, with the magnitude of improvement varying across the evaluated Speech LLMs. Complete Residual Effect Ratio (RER) values for all evaluated models are reported in Table 5 (Appendix F).

5 Conclusion

We introduced AIP-Speech, an inference-only benchmark for evaluating Speech LLM robustness under controlled real-world background interference. By pairing clean and noisy utterances, annotating backgrounds with interpretable acoustic descriptors, and evaluating ASR, keyword spotting, demographic subgroups, and prompt-based steering, AIP-Speech provides a systematic framework for analyzing acoustic interference. Our results show that simple spectral descriptors are strongly associated with degradation, robustness varies substantially across accent groups, and inference-time prompting yields limited improvements.

Limitations

This work is limited to inference-time evaluation of five open Speech LLMs on automatic speech recognition and keyword spotting, and our findings may not generalize to conversational or reasoning-intensive speech tasks. Although the benchmark covers diverse real-world background recordings, it does not exhaust the full range of acoustic environments encountered in practice. The fairness analysis is limited to English accent subgroups and relies on the demographic annotations provided by the underlying datasets. Finally, descriptor-level analysis identifies statistical associations with robustness degradation but does not establish causal relationships between acoustic properties and model behavior.

References

- Abdelrahman Abouelenin, Atabak Ashfaq, Adam Atkinson, Hany Awadalla, Nguyen Bach, Jianmin Bao, Alon Benhaim, Martin Cai, Vishrav Chaudhary, Congcong Chen, et al. 2025. Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras. *arXiv preprint arXiv:2503.01743*.
- Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber. 2020. Common voice: A massively-multilingual speech corpus. In *Proceedings of the twelfth language resources and evaluation conference*, pages 4218–4222.
- Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T Tan, and Haizhou Li. 2026. Voicebench: Benchmarking llm-based voice assistants. *Transactions of the Association for Computational Linguistics*, 14:378–398.
- Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, et al. 2024. Qwen2-audio technical report. *arXiv preprint arXiv:2407.10759*.
- Gemma Team. 2025. [Gemma 3n](#). Google. On-device multimodal model with a Universal Speech Model audio encoder.
- Kai Li, Can Shen, Yile Liu, Jirui Han, Kelong Zheng, Xuechao Zou, Lionel Z Wang, Shun Zhang, Xingjian Du, Hanjun Luo, et al. 2025. Audiotrust: Benchmarking the multifaceted trustworthiness of audio large language models. *arXiv preprint arXiv:2505.16211*.
- Hongcheng Liu, Yixuan Hou, Heyang Liu, Yuhao Wang, Yanfeng Wang, and Yu Wang. 2025. Vocalbench-df: A benchmark for evaluating speech llm robustness to disfluency. *arXiv preprint arXiv:2510.15406*.
- Brian McFee, Colin Raffel, Dawen Liang, Daniel PW Ellis, Matt McVicar, Eric Battenberg, Oriol Nieto, et al. 2015. librosa: Audio and music signal analysis in python. *SciPy*, 2015(18-24):7.
- Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. 2015. Librispeech: an asr corpus based on public domain audio books. In *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 5206–5210. IEEE.
- Jing Peng, Yucheng Wang, Bohan Li, Yiwei Guo, Hankun Wang, Yangui Fang, Yu Xi, Haoyu Li, Xu Li, Ke Zhang, et al. 2025. A survey on speech large language models for understanding. *IEEE Journal of Selected Topics in Signal Processing*.
- Karol J Piczak. 2015. Esc: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM international conference on Multimedia*, pages 1015–1018.
- Chandan KA Reddy, Ebrahim Beyrami, Jamie Pool, Ross Cutler, Sriram Srinivasan, and Johannes Gehrke. 2019. A scalable noisy speech dataset and online subjective test framework. *arXiv preprint arXiv:1909.08050*.
- Muhammad Shah, David Solans Noguero, Mikko Heikkilä, Bhiksha Raj, and Nicolas Kourtellis. 2025. Speech robust bench: A robustness benchmark for speech recognition. In *International Conference on Learning Representations*, volume 2025, pages 38625–38651.

David Snyder, Guoguo Chen, and Daniel Povey. 2015. Musan: A music, speech, and noise corpus. *arXiv preprint arXiv:1510.08484*.

Andrew Varga and Herman JM Steeneken. 1993. Assessment for automatic speech recognition: Ii. noisex-92: A database and an experiment to study the effect of additive noise on speech recognition systems. *Speech communication*, 12(3):247–251.

Pete Warden. 2018. Speech commands: A dataset for limited-vocabulary speech recognition. *arXiv preprint arXiv:1804.03209*.

Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Ke qin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. 2025. [Qwen2.5-omni technical report](#). *ArXiv*, abs/2503.20215.

Linhao Zhang, Jian Zhang, Bokai Lei, Chuhan Wu, Aiwei Liu, Wei Jia, and Xiao Zhou. 2025. Wildspeech-bench: Benchmarking end-to-end speechllms in the wild. *arXiv preprint arXiv:2506.21875*.

Background	Source
rain	ESC-50
sea waves	ESC-50
engine	ESC-50
vacuum	ESC-50
footsteps	ESC-50
fire	ESC-50
helicopter	ESC-50
clock tick	ESC-50
keyboard	ESC-50
dog	ESC-50
rooster	ESC-50
wind	ESC-50
cafeteria	MS-SNSD
restaurant	MS-SNSD
public square	MS-SNSD
airport	MS-SNSD
air conditioner	MS-SNSD
babble	NOISEX-92
music	MUSAN
hubbub	MUSAN

Table 1: Background recordings used throughout the benchmark.

A Background Noise Set

Table 1 lists the 20 background recordings used in AIP-Speech. Recordings are drawn from four publicly available datasets to provide a diverse range of real-world acoustic environments.

B Acoustic Descriptor Definitions

Each background recording is characterized using a set of standard acoustic descriptors computed directly from the waveform. Unless otherwise specified, descriptor values are averaged over the recording after frame-level computation.

- **Zero Crossing Rate (ZCR):** Measures the rate at which the waveform changes sign and serves as a simple indicator of high-frequency activity.
- **Spectral Centroid:** Represents the center of mass of the magnitude spectrum and is commonly interpreted as the perceptual brightness of a sound.
- **Spectral Bandwidth:** Measures the spread of spectral energy around the spectral centroid, indicating how broadly energy is distributed across frequencies.
- **Spectral Rolloff:** The frequency below which 85% of the spectral energy is contained, providing a measure of the effective upper frequency content.

- **Spectral Flux:** Quantifies frame-to-frame changes in the magnitude spectrum and reflects temporal spectral variation.
- **Spectral Flatness:** Measures how noise-like a signal is by comparing the geometric and arithmetic means of the spectrum. Higher values indicate flatter, more noise-like spectra.
- **Spectral Contrast:** Measures the difference between spectral peaks and valleys across frequency bands, capturing harmonic structure and spectral richness.
- **RMS Energy Variance:** Measures the temporal variability of signal energy throughout the recording.
- **Speech Fraction:** Computed using the Silero Voice Activity Detector (VAD) as the proportion of frames classified as speech.

C Per-Model Descriptor Correlations

To assess whether descriptor–degradation relationships are consistent across architectures, Table 2 reports Spearman rank correlations computed independently for each Speech LLM. Degradation is averaged across the evaluated SNR levels before computing the correlations, consistent with the descriptor-level analysis in the main paper.

D Per-Background Acoustic Descriptor Values

Table 3 reports the acoustic descriptor values computed for every background recording used in AIP-Speech. Providing these values enables direct inspection of the descriptor profile associated with each recording and facilitates reproducibility of the descriptor-level analyses presented in the main paper.

E Complete Fairness Results

Table 4 reports the complete subgroup-wise robustness results underlying Figure 5. Values correspond to the mean increase in Word Error Rate (Δ WER) relative to each subgroup’s clean baseline. Reporting the complete statistics enables reproducibility beyond the visualization shown in the main paper.

F Prompt-based Steering

For the prompt-based robustness experiment, each model is evaluated under two inference settings:

Descriptor	Task	Pooled (All 5)	Qwen2-Audio	Qwen2.5-Omni-7B	Qwen2.5-Omni-3B	Phi-4-MM	Gemma 3n
Zero Crossing Rate	ASR	+0.645	+0.711	+0.704	+0.693	+0.672	+0.469
	KWS	+0.723	-0.126*	+0.585	+0.647	+0.710	+0.678
Spectral Centroid	ASR	+0.629	+0.693	+0.597	+0.558	+0.642	+0.451
	KWS	+0.637	-0.322*	+0.501	+0.488	+0.674	+0.567
Spectral Rolloff	ASR	+0.626	+0.687	+0.586	+0.544	+0.669	+0.448
	KWS	+0.619	-0.228*	+0.444	+0.439	+0.656	+0.581
Spectral Bandwidth	ASR	+0.550	+0.621	+0.522	+0.462	+0.636	+0.337
	KWS	+0.544	-0.118*	+0.406	+0.402	+0.576	+0.519
Spectral Flux	ASR	+0.217	+0.248	+0.263	+0.250	+0.192	+0.107
	KWS	+0.272	-0.173*	+0.384	+0.278	+0.277	+0.226
Spectral Flatness	ASR	+0.080	+0.150	+0.029	-0.030	+0.099	+0.177
	KWS	+0.117	-0.243*	+0.025	-0.029	+0.165	+0.038
Spectral Contrast	ASR	-0.005	+0.053	+0.057	+0.087	-0.065	-0.068
	KWS	+0.072	-0.168*	-0.013	+0.075	+0.062	-0.003
RMS Energy Variance	ASR	-0.217	-0.233	-0.247	-0.179	-0.251	-0.096
	KWS	-0.191	+0.311*	-0.156	-0.205	-0.217	-0.173
Silero VAD Speech Fraction	ASR	+0.316	+0.314	+0.406	+0.397	+0.275	+0.254
	KWS	+0.383	+0.017*	+0.457	+0.431	+0.362	+0.369

Table 2: Spearman rank correlation coefficients (r_s) between acoustic descriptors and task degradation. The **Pooled (All 5)** column reports the aggregate correlations used in Figure 3, while the remaining columns report correlations computed independently for each Speech LLM. Degradation is averaged across the evaluated SNR levels before computing the correlations. Asterisk (*) denotes Qwen2-Audio KWS results under the unsteered baseline evaluation protocol.

(i) the default task instruction (baseline), and (ii) a steering prompt that instructs the model to ignore background interference and attend only to the intended foreground speaker. The same steering prompt is applied consistently across all evaluated Speech LLMs.

F.1 Automatic Speech Recognition (ASR)

Baseline Prompt

Transcribe the speech in this audio clip exactly as spoken. Return only the transcription text.

Steering Prompt

*You are an automatic speech recognition (ASR) system operating in noisy real-world environments. Transcribe **only** the speech spoken by the intended foreground speaker. Ignore background voices, overlapping conversations, music, television, announcements, and all environmental sounds.*

If any foreground speech is unintelligible because of background interference, omit it rather than guessing or hallucinating.

Return only the final transcript without explanations, confidence scores, or additional text.

F.2 Keyword Spotting (KWS)

Baseline Prompt

*Listen to this audio clip. Respond with **only** the spoken word (one word). If no clear word is spoken, respond with SILENCE.*

Steering Prompt

Identify the keyword spoken by the intended foreground speaker. Ignore all background voices, music, television, announcements, and environmental sounds.

If no foreground keyword can be identified, respond with SILENCE.

Return only the keyword or SILENCE without any additional text.

F.3 Prompt Steering Results

Table 5 reports the Residual Effect Ratio (RER) for each evaluated Speech LLM. Values below one indicate that prompt steering reduces the degradation caused by background interference.

G Performance Across Signal-to-Noise Ratios

To evaluate robustness across different interference levels, we report model performance under clean conditions together with additive background interference at 10, 5, and 0 dB SNR.

Background	ZCR	Centroid	Bandwidth	Rolloff	Speech	Flux	Flatness	Contrast	RMS Var.
esc_wind	0.0226	669.76	1330.72	1190.62	0.0000	0.9022	0.0011	17.2040	0.000658
esc_rooster	0.0822	841.66	606.18	1153.43	0.0000	0.6738	0.4599	27.7246	0.001060
esc_dog	0.0161	218.08	142.86	342.65	0.0000	0.3479	0.9091	16.0911	0.000359
esc_clock	0.0288	746.37	1351.25	1076.89	0.0021	1.7850	0.0013	19.2353	0.001402
snsd_airconditioner	0.0333	851.80	1375.42	1699.16	0.0000	0.9540	0.0032	15.7306	0.000312
esc_engine	0.0488	1314.33	1544.96	2801.37	0.0000	0.8921	0.0088	18.0697	0.002576
esc_fire	0.0912	2264.88	2236.70	5048.61	0.0000	2.1282	0.0931	16.1431	0.001323
snsd_square	0.0332	1033.03	1645.28	2351.40	0.1014	1.1128	0.0049	15.2164	0.004249
musan_music	0.1081	1701.98	1668.04	3462.07	0.3362	1.3270	0.0215	20.7004	0.002288
esc_keyboard	0.0848	2388.24	1807.72	4160.08	0.0000	3.3657	0.1094	17.3824	0.000738
esc_footsteps	0.1987	2212.21	1959.91	4548.02	0.0000	1.6759	0.1182	16.8711	0.001814
esc_vacuum	0.5089	2755.58	2109.97	5039.44	0.0000	0.9197	0.0533	19.2603	0.000078
esc_helicopter	0.2609	3211.55	2666.81	6452.88	0.0000	0.9847	0.0935	18.6647	0.000018
snsd_airport	0.1516	1354.49	766.39	2120.14	0.0000	1.1973	0.0002	20.3577	0.001119
snsd_cafeteria	0.0744	1446.31	1905.25	3401.42	0.8367	1.1776	0.0211	15.6066	0.000375
noisex_babble	0.0870	1255.53	1428.23	2535.08	0.4696	1.1716	0.0114	18.5808	0.000327
esc_seawave	0.1446	1859.74	1834.55	3791.06	0.0000	0.8943	0.0655	17.5634	0.000899
esc_rain	0.3420	3129.37	1921.83	5528.74	0.0000	0.9045	0.2122	18.5032	0.000029
snsd_restaurant	0.1472	2110.20	2106.19	4668.49	0.0171	1.0586	0.0909	14.9841	0.000437
musan_hubbub	0.1406	1904.30	1705.12	3701.78	0.6201	1.9962	0.0648	19.4705	0.002083

Table 3: Per-recording acoustic descriptor values for every background recording used in AIP-Speech. These descriptor profiles underlie the descriptor-level correlation analysis presented in the main paper and provide a recording-level acoustic characterization for every background recording.

Accent Group	Gemma 3n	Phi-4-MM	Qwen2.5-Omni-3B	Qwen2.5-Omni-7B	Qwen2-Audio
Australian English	-0.1316	+0.0924	+0.1309	+0.0906	+0.0681
Canadian English	-0.2173	-0.0915	-0.2280	+0.0866	+0.3350
England English	-0.2835	+0.0542	+0.0705	+0.0490	+0.0888
Hong Kong English	+0.8583	+0.0583	+0.4083	+0.0625	-0.4125
Indian English	+0.5370	+0.1628	+0.2425	+0.1967	+0.1713
Irish English	+0.2583	+0.0461	+0.0500	+0.0356	+0.2628
New Zealand English	+0.1071	0.0000	+0.0143	+0.0238	+0.6881
Scottish English	+0.2211	+0.2164	-0.4461	+0.0261	+0.1708
US English	+0.0854	+0.0480	+0.0579	+0.0522	+0.0490
Other / Misc.	+0.2470	+0.1526	+0.1425	+0.1453	+0.1915

Table 4: Complete subgroup-wise robustness statistics (mean Δ WER) used in the fairness analysis. Larger positive values indicate greater performance degradation caused by background interference relative to clean baseline.

Model	ASR	KWS
Gemma 3n	1.664	0.767
Phi-4-MM	2.423	0.549
Qwen2.5-Omni-3B	0.812	2.489
Qwen2.5-Omni-7B	1.064	1.681
Qwen2-Audio	5.989	1.000

Table 5: Residual Effect Ratio (RER) after prompt-based steering. Values below one indicate reduced degradation relative to the baseline prompt.

Figure 7 summarizes the average performance of all evaluated Speech LLMs across the two benchmark tasks. As expected, lower signal-to-noise ratios consistently degrade model performance. The figure provides additional context for the descriptor-level analysis presented in the main paper by illustrating the overall effect of increasing acoustic interference.

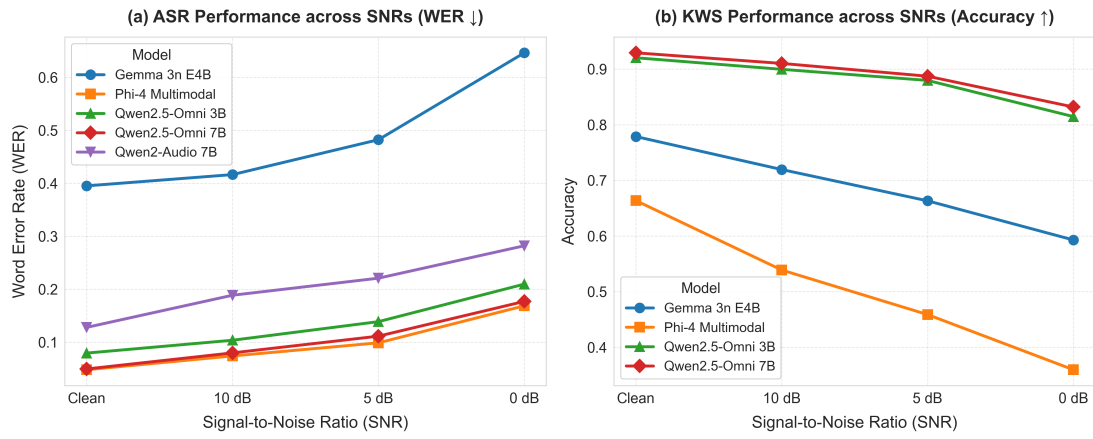


Figure 7: Average model performance under clean conditions and progressively more challenging acoustic environments (10, 5, and 0 dB SNR). (a) Automatic Speech Recognition performance measured using Word Error Rate (WER; lower is better). (b) Keyword Spotting performance measured using classification accuracy (higher is better). Performance consistently deteriorates as background interference increases, motivating the descriptor-level robustness analysis presented in the main paper.